

IMU-Based Crash Detection

& Driver Behaviour Analytics · Vecmocon IoT Platform

A safety-intelligence layer that converts raw 6-axis inertial sensor data into crash alerts, harsh-driving events, and per-driver risk scores — unlocking new revenue streams in insurance telematics and fleet safety while reducing operational risk for EV fleet operators.

↑ Revenue

New telematics/insurance line

↓ Risk

Faster incident response

↑ Retention

Stickier fleet customers

Real-time

On-device detection

01. WHY WE NEED THIS PROJECT

Vecmocon's IoT platform powers electric two- and three-wheeler fleets across India. Before this project, the platform tracked location and battery health but had no awareness of how a vehicle was being driven or whether it had crashed. This was a strategic gap for three reasons:

- ▶ **Safety blind spot:** A crash could go undetected for a long time, delaying emergency response and exposing fleet operators to liability and reputational damage.
- ▶ **Untapped data value:** The IMU sensor was already present in the hardware, but its data was unused. Competitors were beginning to offer driver-scoring and crash-detection as premium features.
- ▶ **No behavioural lever:** Fleet operators had no objective way to identify risky drivers, coach them, or reward safe ones — a direct driver of accident rates, vehicle wear, and insurance cost.

Core thesis: *The hardware to detect crashes and risky driving already existed in the field. The opportunity was a pure software/analytics play — high margin, fast to deploy over-the-air, and immediately monetisable.*

02. BUSINESS & REVENUE IMPACT

Direct revenue opportunities

- ▶ Insurance telematics: Objective crash and driving data enables usage-based-insurance (UBI) partnerships, where insurers pay for verified risk data — a recurring, high-margin revenue stream.
- ▶ Premium safety tier: Crash detection and driver scoring can be packaged as a paid add-on above the base IoT subscription, lifting average revenue per vehicle.
- ▶ Fleet-safety SaaS: Driver leaderboards, coaching reports, and risk dashboards become a sellable module to large fleet operators.

Indirect / defensive value

Impact Area	Mechanism	Business Outcome
Customer retention	Safety features raise switching cost	Lower churn on the ₹40L/month line
Insurance premiums	Verified safe-driving data	Lower fleet insurance costs → stronger value prop

Impact Area	Mechanism	Business Outcome
Vehicle lifetime	Reduced harsh driving via coaching	Less battery & component wear, lower warranty cost
Brand & trust	Faster crash response	Differentiation vs. location-only competitors

03. CUSTOMER SATISFACTION IMPACT

- ▶ **Fleet operators:** Gain peace of mind and a concrete tool to reduce accidents, manage liability, and lower insurance costs — directly improving their unit economics.
- ▶ **Drivers:** Safe drivers are recognised and rewarded; risky drivers get early, data-backed coaching instead of arbitrary blame.
- ▶ **OEM partners (e.g. TATA AutoComp):** Can market vehicles with built-in safety intelligence, strengthening their own product proposition.

Satisfaction signal: *The feature turns a passive tracking device into an active safety partner — moving the customer relationship from 'utility' to 'indispensable'.*

04. STAKEHOLDER ALIGNMENT

Delivering a safety-critical analytics feature required aligning several groups with competing priorities. As product proxy, I owned this alignment.

Stakeholder	Their Priority	How I Aligned Them
Engineering / Embedded	Device compute & power limits	Agreed on-device event detection over raw-data streaming to fit constraints
Fleet customers	Trustworthy, low-false-alarm alerts	Co-defined acceptable false-positive rate before building thresholds
Data / Cloud team	Bandwidth & storage cost	Stream only events + short data windows, not continuous raw IMU
Leadership / Sales	A monetisable, differentiated feature	Framed roadmap around insurance & premium-tier revenue
OEM partners	Reliability & brand safety	Set clear accuracy criteria and phased rollout

I translated these into PRDs and user stories with explicit acceptance criteria, then prioritised which events to ship first (crash detection first, given its safety and insurance value) using a RICE-style assessment of customer impact.

05. HOW IT WORKS — DETECTION FLOW

The feature runs as a layered pipeline on the in-vehicle device, escalating only meaningful events to the cloud:



Step detail

- ▶ **1 · IMU Raw Data:** 6-axis accelerometer + gyroscope continuously sampled on the device.
- ▶ **2 · On-Device Filtering:** Noise and road-artefact rejection (potholes, bumps) using duration, multi-axis correlation, and post-event motion checks.
- ▶ **3 · Event Classification:** Signatures matched to events: crash, harsh braking, rapid acceleration, sharp cornering — with severity scoring via jerk and deceleration profiles.
- ▶ **4 · Cloud + Alert:** Only classified events + a short data window are sent up; crash events trigger immediate alerts; behaviour events aggregate into a per-driver score.

Driver scoring logic (conceptual)

- ▶ Each detected event is weighted by type and severity.
- ▶ Events are normalised against vehicle-class baselines (a loaded e-rickshaw ≠ a light scooter).
- ▶ Aggregated into a rolling driver score that fleet operators can rank, coach, and reward against.

06. OUTCOME & STRATEGIC VALUE

- ▶ Delivered a safety-critical, real-time detection feature within device compute and bandwidth limits.

- ▶ Created the foundation for a new insurance-telematics revenue stream and a premium safety tier.
- ▶ Strengthened retention on a ₹40L/month product line by deepening the platform's value to fleet customers including TATA AutoComp.
- ▶ Demonstrated the platform's ability to extract monetisable value from sensors already deployed in the field — a repeatable strategic pattern.

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